



Cohort 3.0 Test (19-8-26)

1 What happens when you enter `https://example.com` in a browser?

☐ The browser directly executes the website's JavaScript

☒ DNS resolves the domain, then the browser communicates with the server

Correct Selection

☐ The browser converts the domain directly into HTML

☐ The browser creates a new IP address

2 Which HTTP status code indicates that a request was successful?

☐ 201

☐ 301

☒ 200

Correct Selection

☐ 404

3 What is the primary difference between HTTP and HTTPS?

☐ HTTPS is faster than HTTP

☒ HTTPS encrypts communication using TLS

Correct Selection

☐ HTTP does not use IP addresses

☐ HTTPS is only used for images

4 Why does React use a virtual DOM?

☐ To replace JavaScript

☐ To store application data permanently

☒ To efficiently update the actual DOM

Correct Selection

☐ To communicate directly with databases

5 What is the purpose of the key prop when rendering a list in React?

☒ To uniquely identify elements during reconciliation

Correct Selection

☐ To encrypt list data

☐ To create CSS classes

☐ To store component state

6 What happens when a React component's state changes?

☐ The browser always reloads the entire page

☐ The component is permanently deleted

☐ The server automatically restarts

☒ React schedules a re-render of the component

Correct Selection

7 What is the main advantage of client-side routing in a React SPA?

☒ It allows navigation without a full browser page reload

Correct Selection

☐ It eliminates JavaScript

☐ It requires a new HTML page for every route

☐ It prevents API requests

8 What should a protected route typically check before rendering a private page?

☐ Screen size

☐ Browser color scheme

☒ Authentication/authorization state

Correct Selection

☐ CSS file size

9 What will count typically become after the update?
`const [count, setCount] = useState(0);`

`setCount(count + 1);`
`setCount(count + 1);`

☒ 1

Correct Selection

☐ 2

☐ 0

☐ 3

10 Which approach correctly performs two sequential state updates based on the previous state?

☐ `setCount(count + 1); setCount(count + 1);`

☐ `setCount(count++); setCount(count++);`

☒ `setCount(prev => prev + 1); setCount(prev => prev + 1);`

Correct Selection

☐ `count = count + 2;`

11 What does an empty dependency array in `useEffect` generally indicate?

- ☐ The effect runs after every render
- ☐ The effect does not run
- ☐ The effect runs only when props change
- ☒ The effect runs after the initial mount

Correct Selection

12 Why is a cleanup function used inside `useEffect`?

- ☒ To clean up subscriptions, timers, event listeners, etc.
- ☐ To remove unnecessary CSS
- ☐ To create a new component
- ☐ To update the URL

Correct Selection

13 What problem does Context API primarily help solve?

- ☒ Prop drilling
- ☐ Memory leaks
- ☐ HTTP encryption
- ☐ Database normalization

Correct Selection

14 If a component calls `useContext(MyContext)`, what does it receive?

- ☐ The entire React application
- ☒ The current value provided by the nearest matching Provider
- ☐ Only the Provider component
- ☐ The browser's local storage

Correct Selection

15 What is the primary purpose of `React.memo()`?

- ☒ To memoize a component and potentially skip unnecessary re-renders
- ☐ To store data in a database
- ☐ To memoize API responses automatically
- ☐ To create global state

Correct Selection

16 When should `useMemo()` generally be used?

- ☐ For every variable in a component
- ☐ Instead of `useState()` in every situation

☐ To replace all `useEffect()` calls

☒ When memoizing a meaningful expensive computation can improve performance

Correct Selection

17 What does `useCallback()` primarily memoize?

☐ A component

☐ An API response

☒ A function reference

Correct Selection

☐ A DOM element

18 In Redux, which statement best describes a reducer?

☒ It calculates the next state from the current state and an action

Correct Selection

☐ It sends HTTP requests automatically

☐ It directly modifies the existing state object

☐ It renders React components

19 What is the purpose of `useSelector()` in React-Redux?

☐ To dispatch an action

☐ To create a reducer

☒ To read data from the Redux store

Correct Selection

☐ To create a new store

20 What is the purpose of `useDispatch()`?

☐ To read CSS properties

☒ To access the dispatch function and send actions to the Redux store

Correct Selection

☐ To create React components

☐ To fetch HTML files

21 Why is Node.js well suited for handling many I/O operations?

☐ It creates a new CPU thread for every request

☐ It does not use JavaScript

☐ It stores all requests permanently in memory

☒ It uses an event-driven, non-blocking I/O model

Correct Selection

22 What is the purpose of package.json in a Node.js project?

☐ It stores only HTML code

☒ It contains project metadata, scripts, and dependency information

Correct Selection

☐ It stores the operating system

☐ It replaces JavaScript files

23 Which statement about asynchronous operations in Node.js is correct?

☒ Node.js can continue handling other work while waiting for asynchronous I/O

Correct Selection

☐ Every asynchronous operation blocks the entire application

☐ Node.js cannot handle multiple requests

☐ Async operations are only possible in browsers

24 A React frontend needs data from a Node.js backend. What is the most common communication method?

☐ CSS

☒ HTTP API requests

Correct Selection

☐ JSX compilation

☐ React Context only

25 A user logs in successfully and then tries to access /dashboard. What is the best general approach for protected routing?

☐ Always render the dashboard regardless of login status

☐ Hide the dashboard using CSS only

☐ Store the password directly in the URL

☒ Check authentication state and redirect unauthenticated users

Correct Selection

26 What happens if you directly modify a state variable instead of using its setter?

☐ The browser reloads

☐ The component is automatically deleted

☒ React may not re-render the component

Correct Selection

☐ React always detects the change

27 What happens when the dependency array contains a variable?

```
useEffect(() => {  
  console.log("Hello");  
}, [count]);
```

☒ The effect runs whenever count changes

Correct Selection

- ☐ The effect runs only once
- ☐ The effect never runs
- ☐ The effect runs only when the component is deleted

28 What happens when the value of a Context Provider changes?

- ☒ Consumers using that context can re-render Correct Selection
- ☐ The entire browser reloads
- ☐ The context is automatically deleted
- ☐ Redux state is changed

29 Which hook is most appropriate for memoizing a function reference?

- ☐ useState()
- ☐ useEffect()
- ☒ useCallback() Correct Selection
- ☐ useContext()

30 Which sequence best describes a typical Redux state update?

- ☐ Reducer → Action → Dispatch → Store
- ☐ Store → Component → Action → Reducer
- ☐ Dispatch → Component → CSS → Store
- ☒ Action → Dispatch → Reducer → Store update Correct Selection

31 Why should Redux state generally be treated as immutable?

- ☐ To prevent React from running
- ☒ To make state changes predictable and enable efficient change detection Correct Selection
- ☐ To increase API response time
- ☐ To prevent routing

32 What does the Node.js event loop primarily help with?

- ☒ Handling asynchronous operations without blocking the main JavaScript thread Correct Selection
- ☐ Creating HTML files
- ☐ Compiling React components
- ☐ Managing CSS animations

33 Which HTTP method is generally used to create a new resource?

☐ GET

☐ DELETE

☒ POST

Correct Selection

☐ PATCH

34 A user is not authenticated and tries to access /admin. What should a protected route typically do?

☐ Show the admin page

☒ Redirect the user to a login page

Correct Selection

☐ Delete the user's account

☐ Restart the server

35 What does a 404 Not Found HTTP status usually mean?

☐ The server successfully processed the request

☐ The server is permanently offline

☐ The request was successfully authenticated

☒ The requested resource could not be found

Correct Selection

Close